

THE EFFECTS OF PRIVACY REGULATION ON THE MARKET FOR STOLEN DATA ONLINE APPENDIX

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1 Data

1.1 UK Survey Data

The UK survey data referenced throughout the paper are from the United Kingdom Cyber Security Breach Survey: Combined Dataset, 2016-2022 ([Department for Digital, Culture, Media and Sport 2022](#)). I accessed the data through the UK Data Services online portal on March 20, 2023.

Only the 2018 and 2019 surveys asked respondents whether they made any changes in response to the GDPR. The survey asked about the types of changes made as well, which I have combined into five groups: human changes (e.g., staff training and hiring), technical changes (e.g., updated system configurations and increased spending on security), policy changes (e.g., conducting more audits and changing who has admin rights), third-party changes (e.g., changing IT service providers), and other changes (e.g., changing the nature of the business).

For figure [A1](#), an organization was considered breached if they reported a ransomware or other malware infection; hacking of bank accounts; phishing attacks; unauthorized file access; or any other breach or attack.

1.2 Breach Data

The individual breach data obtained for this study contains many more data package observations than are included in the final paper. Observations were dropped for one of three reasons. First, any breaches that could not be attributed to an organization or country were removed. Second, any data package that was discovered during a breach of a breach-indexing website, or similar “breach of breaches” was dropped. These breaches are of websites and other platforms that bundle access to credentials leaked in other breaches to their users. Essentially, the data being leaked in those breaches had itself been stolen from its original owner. What makes these observations unusable is the lack of a clear date when the data were originally stolen. The observed date is of the larger breach, but it is unknown when the

smaller breaches that comprise the breach occurred. Finally, data packages that appeared online prior to 2017 were removed. As briefly discussed when the panel data was described, the organization collecting these data was founded in 2016. Dropping these early breaches allows for the possibility that the breaches collected prior to that founding were meaningfully different from those that were collected later.

2 Defining Personally Identifiable Information

From a legal standpoint, there are three commonly used definitions of “personally identifiable information” (Schwartz and Solove 2011). The tautological definition used in the Video Privacy Protection Act says that PII is information which identifies a person. The non-public information approach used in the Gramm-Leach-Bliley Act defines PII as non-public personal information. Finally, the specific-types approach explicitly lists the types of data that are considered PII. I borrow from all three approaches.

In the data I am able to observe the specific types of records in the data packages. I classify data as PII if it reveals location, financial, contact, user account, or personal information. Account information covers emails, usernames, and passwords. Personal information includes political and religious views, sexual orientation, and aspects of a person’s home life such as if they have children or pets. As most of the data packages included emails and passwords (figures 4 and A2), this makes the fraction of records in a data package that are PII fairly close to one. As part of my robustness checks, I repeat the data package analysis of the effect of the GDPR on the fraction of records in a data package that are PII using an alternative definition that removes emails and passwords. I find that this did not change the main result that the GDPR had no effect on the portion of records in a breach that are PII (table A24).

3 Descriptive Information

Tables A2-A5 report unconditional differences in means between various data package groups.

Table A2 compares treated and untreated data packages across the full sample. There are statistically significant differences between the two in the fraction of records that are PII, and the number of unique data types. Although they are statistically significant, they are not particularly meaningful. Given that both types have close to 70 percent PII, a 4 percentage point difference is not particularly large. And the difference in number of unique data types is less than one, making them effectively the same from an interpretation perspective.

Data packages that became available before and after the GDPR are then compared in table A3. The data packages get significantly larger after the GDPR in terms of both the number of records and the number of unique data types. As shown in tables A4 and A5, which compare the packages pre- and post-GDPR for the control and treated groups, respectively, this effect is seen in both, though it is much larger in the treated group. This is consistent with the findings that expected data package size significantly increased after the GDPR, and the theory that attackers may have shifted their efforts towards larger targets.

3.1 Dropped Observations

The observations dropped from the final datasets because either their country of origin or breach date could not be determined tend to contain fewer records than those included in the study (figure A3). There are three periods where a large number of breaches were dropped: the first and second quarters of 2018, and the fourth quarter of 2020. In each of these periods there was a data breach whose contents were an amalgamation of data from many other smaller breaches. The 2020 breach specifically, known as the Cit0Day breach, was a collection of over 23,000 breached websites bundled together. The Cit0Day website collected each of those smaller breaches and offered access to the information they contained for a fee. These observations are dropped because it is not possible to identify when these smaller breaches occurred. It is possible they were breaches that occurred years prior to the larger breach, or right before. Figure A4 plots the number of breaches and records included and excluded from the final sample over time.

4 Results

Additional results from alternative specifications of the estimated models are presented here.

4.1 Extensive Margin Effects

On the extensive margin, I separately estimate equation 5.1 for small and large countries. The former are countries with populations below the median population in 2018, the latter countries with populations above the median population in 2018. Results are in tables A6 and A7.

4.2 Aggregate Effects

To test the robustness of my aggregate effect estimates, I re-estimate each aggregate effect after removing a treated country from the data. For each removed country, the estimate stays well within the 95 percent confidence interval of the estimate with the full sample (figures A5 and A6).

Next, I use different methods to construct the panel. Brazil and China each adopted their own data privacy laws near the end of the study period. Removing them from the sample slightly reduces the estimated effect on the number of breaches, though it is still significant. As before, there are no statistically significant effects on the number of records available (table A8).

Table A9 shows the aggregate results when I exclude data from after the first quarter of 2020 to avoid any pandemic effects. This significantly reduces the number of post-treatment observations. The change in the number of records remains insignificant and in number of data breaches significant, but the long-run effect in the latter case does change. The short-run effects on both outcomes are identical to using the full panel, which is unsurprising since observations in the pre-treatment period and short-run all remain in this new panel. The only change is in the long-run estimates, where the reduction in number of data breaches shrank, though is still statistically significant. In this shorter panel there are only three

long-run periods: the third and fourth quarters of 2019 and the first quarter of 2020. These results suggest that the long-run effect grows as time goes on.

Table A10 shows the quantity results when I exclude data packages originating in multinational organizations from the panel. Whether an organization is a multinational is determined in one of two ways. First, if their website is hosted in more than one country, they are considered multinational. Or, they are considered multinational if their website and organizational information, such as privacy policies, discuss having customers or users in more than one country. The argument for excluding these effects is that multinational organizations may be partially treated. The GDPR applies to data specifically from EU residents. A multinational organization would therefore have to comply if they have any users in the EU, but it is not clear whether they would change their data collection and protection practices for all their users, or just those in the EU.

When multinational breaches are excluded, there is actually a long-run increase in the number of records available after the GDPR. The effect on the number of data breaches is roughly equivalent to the one found in the main specification.

Each of the previous tests left the definition of the outcome variables unchanged and were estimated with the same Poisson regression as in the main paper. Tables A11-A16 test changes in the outcome variable definition, the effect of adding covariates to the equation, and using three other models to derive estimates.

First, I estimate the effect using a linear model, rather than a Poisson model:

$$Y_{it} = \gamma_s + \tau_t + \delta D_i \times Post-GDPR_t + \epsilon_{it} \quad (1)$$

where each term is defined as before. In addition to using a linear model, I use two log-like transformations of the outcome variable, $\log(Y_{it} + 1)$ and the inverse hyperbolic sine (IHS) function $\ln(Y_{it} + \sqrt{Y_{it}^2 + 1})$. These transformations are necessary, rather than just using $\log(Y_{it})$ because there are a number of periods in which countries have no breaches. Using these transformations significantly changes the results from the Poisson model. For the number of records, both the $\log(Y_{it} + 1)$ and IHS transformation give large and statistically significant negative estimates of the treatment effect, unlike the Poisson which showed no change. I believe this is due to a significant extensive margin effect. Chen and Roth 2023 and Mullahy and Norton 2024 both discuss how, when there are mass points at zero, log-like transformations may greatly influence the estimated coefficients.

As discussed in the main body of the paper, there are significant and negative extensive margin effects (table 4). This is likely the source of the discrepancies in effect sizes between the models and the primary reason for using the Poisson model over the linear models with a log-like transformation.

Without the log-like transformation, when it is estimated in levels, the linear model produces results that are in line with, though interpreted differently than, the Poisson model. Specifically, I still find no effect on the number of records and a significant negative effect on the number of data breaches (columns 7 and 8 of each table).

Next, I estimate the models using various measures to account for population size. In tables A13 and A14, I add population in millions as a covariate. It is not included in the levels models because the outcomes are already scaled to be records/data packages per million. In all cases there is no significant change in the estimates and the population coefficient is

insignificant.

In tables A15 and A16, I change the outcome for the log-like transformation to also be number of records/data packages per million, and add a population offset to the Poisson model. This noticeably changes the magnitude of both log-like transformations in each outcome. As Chen and Roth 2023 discuss, this is a reflection of the sensitivity of log-like transformations to the scale of the outcome variable when extensive margin effects are present. The offset in the Poisson model effectively changes the outcome to a rate, as in breaches per million. The estimates, however, are roughly the same as those in the model without the offset.

For my final model specification tests, I used alternative ways of controlling for population and added covariates. My results were largely unchanged in each case. In table A17, I remove the population offset. Table A18 weights the estimates using population rather than including an offset. In table A19, I again remove the population offset and opt instead for using per capita outcome variables. Finally, I split the sample into small and large countries in tables A20 and A21, and include indicators for whether the observation is a small or large country in table A22.¹

A shortcoming of my data is that many variables that would be reasonable to include as covariates, such as the fraction of people with internet access, are not consistently observed for every country. Rather than drop observations and unbalance the panel to account for this, the only covariate I add to the model is GDP per capita. As shown in table A23, this does not have a significant effect on my effect estimates.

4.3 Data Package Effects

Estimates of the change in PII fraction using a slightly different definition of PII are in table A24. Under this definition, I remove emails and passwords from PII. I find no significant change, as is the case using the original definition.

As previously discussed, data packages from periods prior to January 2017 were excluded from the main dataset. Tables A26-A28 report the results using the full sample, including those early breaches. In all cases, the signs of the estimated coefficients remain the same. The magnitude of the increase in number of records is larger (comparing table A26 to table ??), but the estimates are still within each other’s standard errors. For the number of unique data types, using the full sample does result in a statistically significant increase, unlike the smaller sample. But the increase is still less than a single data type and therefore not economically meaningful.

Finally, I estimated extensive margin effects for each of the data types using the linear probability model

$$Positive_i = \gamma_i + \tau_t + \delta D_{it} + \varepsilon_{it}$$

where $Positive_i$ is one if the data package contains a positive amount of that data; γ_i and τ_t are country and quarter fixed effects, respectively; and D_{it} is an indicator for whether the data package is treated.

There is a short-run increase in the probability of a data package containing email addresses and password information, but neither is maintained into the long run. Long term,

1. Small or large in this context means above or below the median population in 2018.

the only data type showing a significant change is account information, which saw an eight percent increase in the likelihood that it is in a data package (table [A29](#)).

References

- Chen, Jiafeng, and Jonathan Roth. 2023. “Logs with Zeros? Some Problems and Solutions*.” *The Quarterly Journal of Economics* 139, no. 2 (December): 891–936. ISSN: 0033-5533. <https://doi.org/10.1093/qje/qjad054>.
- Department for Digital, Culture, Media and Sport. 2022. *Cyber Security Breaches Survey: Combined Dataset, 2016-2022*. data collection. SN: 8971, DOI: <http://doi.org/10.5255/UKDA-SN-8971-1>.
- Mullahy, John, and Edward C. Norton. 2024. “Why Transform Y? The Pitfalls of Transformed Regressions with a Mass at Zero” [in en]. *Oxford Bulletin of Economics and Statistics* 86 (2): 417–447. ISSN: 1468-0084. <https://doi.org/10.1111/obes.12583>.
- Schwartz, Paul M., and Daniel J. Solove. 2011. “The PII Problem: Privacy and a New Concept of Personally Identifiable Information” [in eng]. *New York University Law Review* 86 (6): 1814–1894.

Online Appendix Figures

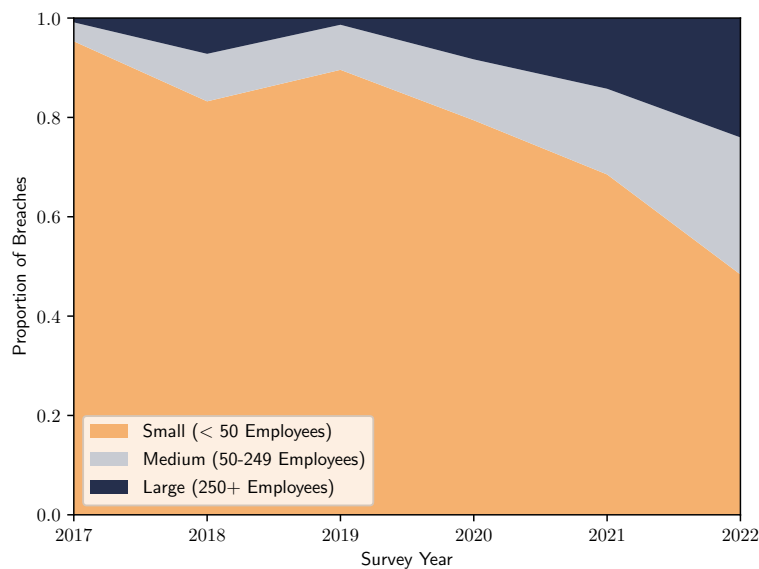
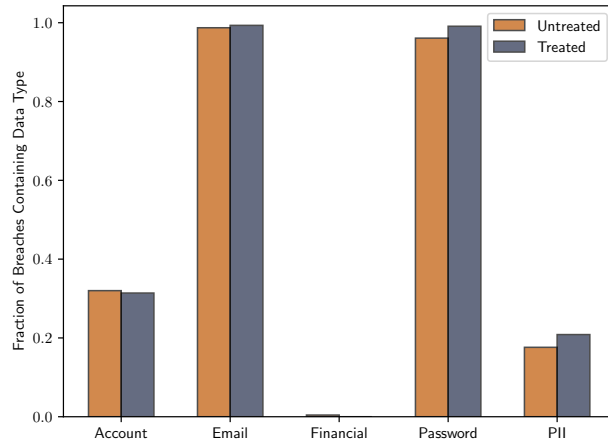
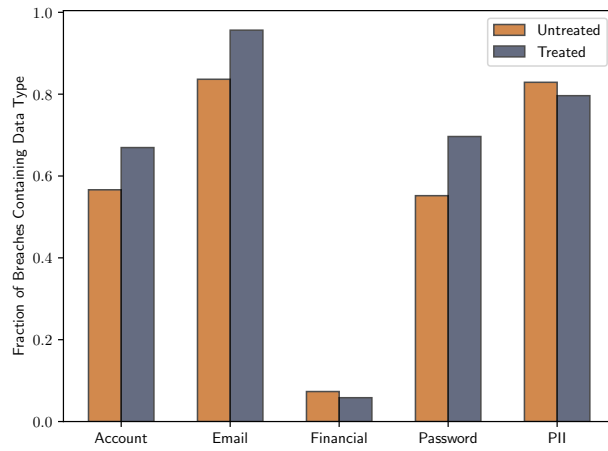


Figure A1: Portion of Data Breaches by Size

Note: This figure shows the proportion of *Source:* [Department for Digital, Culture, Media and Sport \(2022\)](#), author’s calculations.



(a) Pre-GDPR



(b) Post-GDPR

Figure A2: Fraction of Data Packages Containing Each Data Type, Pre- and Post-GDPR

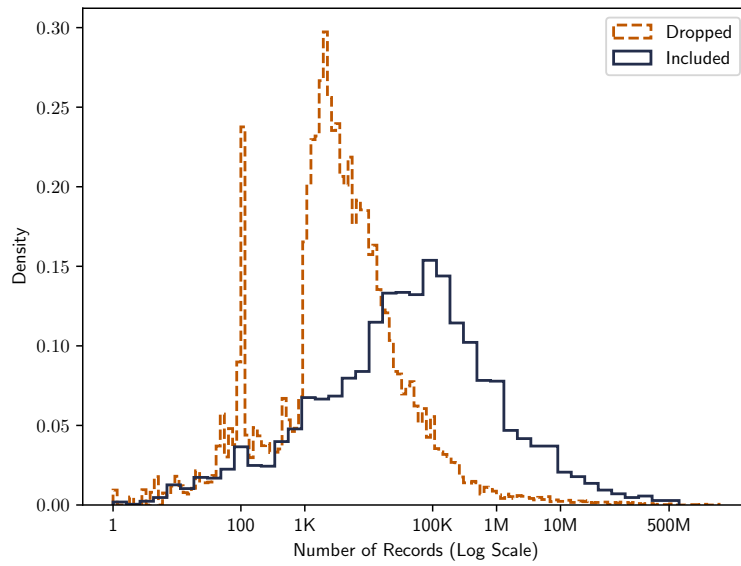
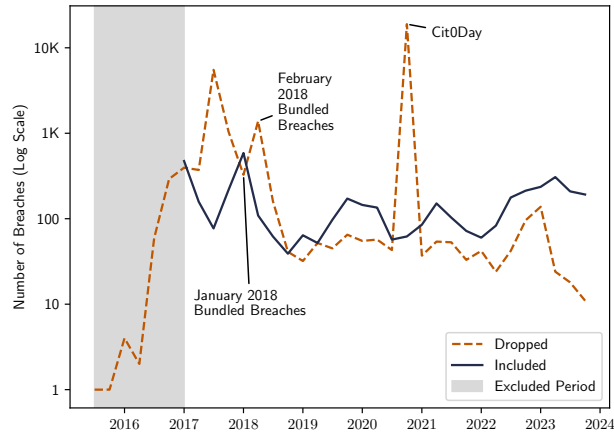
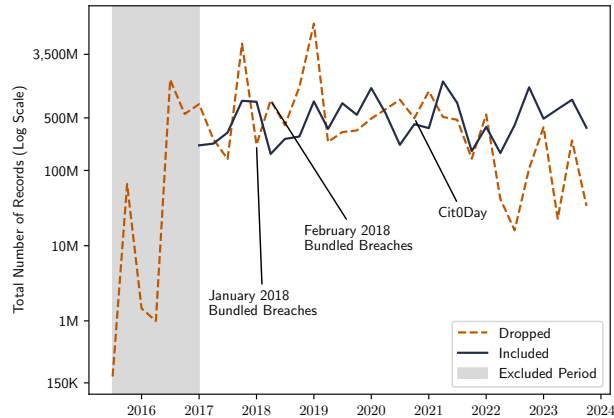


Figure A3: Distribution of Records Per Breach: Dropped vs. Included

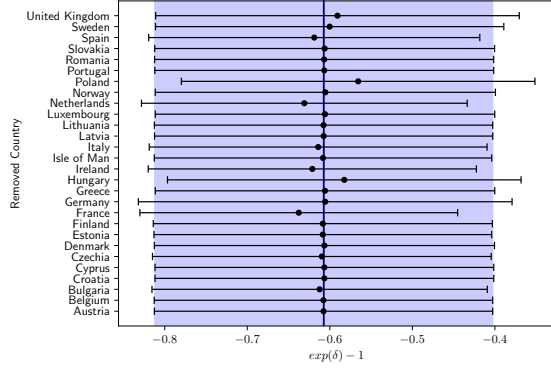


(a) Number of Breaches

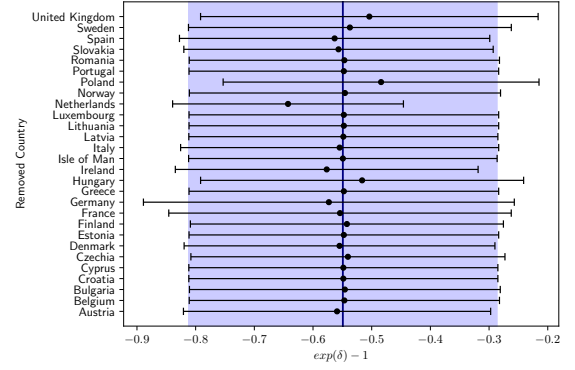


(b) Number of Records

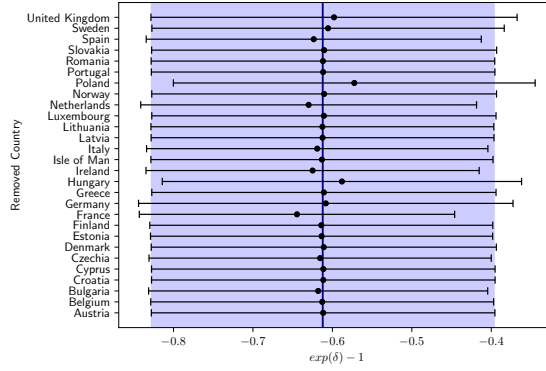
Figure A4: Comparison of Dropped and Included Breaches Over Time



(a) Overall Effect



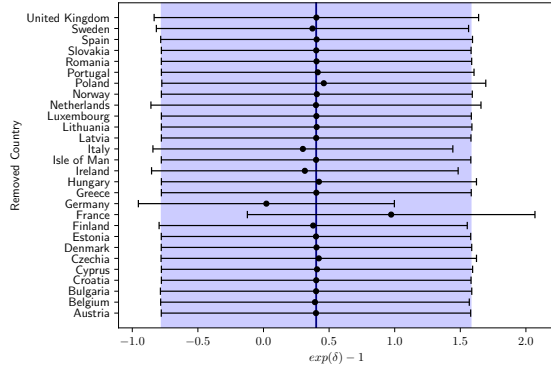
(b) Short Run Effect



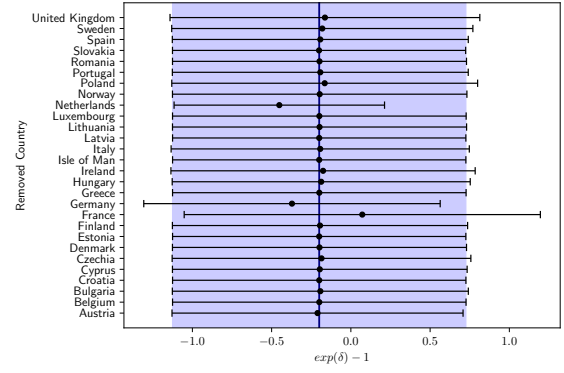
(c) Long Run Effect

Figure A5: Number of Data Breaches Effects Removing Countries

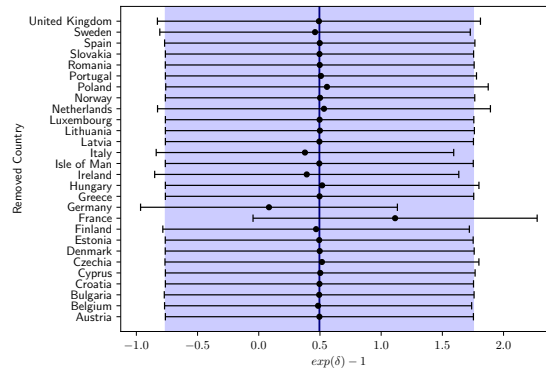
Note: Each point represents the estimated effect on number of data breaches after removing observations from the specified country. The whiskers are the 95 percent confidence interval. The solid line is the point estimate including all countries, and the shaded area is the 95 percent confidence interval around that point.



(a) Overall Effect



(b) Short Run Effect



(c) Long Run Effect

Figure A6: Number of Records Effects Removing Countries

Note: Each point represents the estimated effect on number of records after removing observations from the specified country. The whiskers are the 95 percent confidence interval. The solid line is the point estimate including all countries, and the shaded area is the 95 percent confidence interval around that point.

Online Appendix Tables

Table A1: UK Cyber Security Breach Survey Dates and Sample

Survey Year	Sample Size	Survey Period
2016	1,008 businesses	November 30, 2015 – February 5, 2016
2017	1,523 businesses	October 24, 2016 – January 11, 2017
2018	1,519 businesses, 569 charities	October 9, 2017 – December 14, 2017
2019	1,566 businesses, 514 charities	October 10, 2018 – December 20, 2018
2020	1,348 businesses, 337 charities	October 9, 2019 – December 23, 2019
2021	1,419 businesses, 487 charities, 378 educational institutions	October 12, 2020 – January 21, 2021
2022	1,243 businesses, 424 charities, 490 educational institutions	September 20, 2021 – January 21, 2022

Note: Survey data and documentation can be accessed via the UK Data Service digital catalogue.

Table A2: Data Package Means: Treated vs. Untreated

	Means		Overall Mean N=4,394	Differences
	0 N=3,467	1 N=927		Treated - Untreated
Number of Records	3,309,228 (465,094.606)	4,422,933 (1,128,570.651)	3,544,186 (437,434.656)	1,113,705 (1,220,649.297)
PII Fraction	0.700 (0.003)	0.665 (0.006)	0.693 (0.003)	-0.035*** (0.007)
# of Data Types	6.363 (0.089)	5.688 (0.166)	6.220 (0.079)	-0.674*** (0.188)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Data Package Means: Pre-GDPR vs. Post-GDPR

	Means		Overall Mean N=4,394	Differences
	0 N=1,621	1 N=2,773		Post - Pre
Number of Records	1,598,365 (465,366.681)	4,681,646 (636,601.781)	3,544,186 (437,434.656)	3,083,281*** (788,560.699)
PII Fraction	0.551 (0.003)	0.776 (0.003)	0.693 (0.003)	0.225*** (0.004)
# of Data Types	3.163 (0.069)	8.007 (0.104)	6.220 (0.079)	4.844*** (0.125)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Data Package Means: Pre- vs. Post-GDPR, Untreated

	Means		Overall Mean N=3,467	Differences
	0 N=1,175	1 N=2,292		Post - Pre
Number of Records	2,123,526 (640,456.742)	3,917,082 (621,924.394)	3,309,228 (465,094.606)	1,793,556** (892,734.445)
PII Fraction	0.552 (0.004)	0.776 (0.004)	0.700 (0.003)	0.224*** (0.005)
# of Data Types	3.279 (0.090)	7.943 (0.113)	6.363 (0.089)	4.664*** (0.145)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Data Package Means: Pre- vs. Post-GDPR, Treated

	Means		Overall Mean N=927	Differences
	0 N=446	1 N=481		Post - Pre
Number of Records	214,813 (92,543.690)	8,324,849 (2,159,205.555)	4,422,933 (1,128,570.651)	8,110,036*** (2,161,187.859)
PII Fraction	0.547 (0.004)	0.775 (0.008)	0.665 (0.006)	0.228*** (0.009)
# of Data Types	2.859 (0.080)	8.312 (0.259)	5.688 (0.166)	5.453*** (0.271)

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Extensive Margin Effects: Small Countries

	Dependent Variable: Positive Number of Breaches	
	(1)	(2)
Post $\times$ Treatment	-0.213*** (0.049)	
SR $\times$ Treatment		-0.211*** (0.064)
LR $\times$ Treatment		-0.213*** (0.049)
Observations	1,372	1,372
R^2	0.323	0.323
Period Fixed Effects	Y	Y
Country Fixed Effects	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Small countries are defined as those with a population below the median in 2018.

Table A7: Extensive Margin Effects: Large Countries

	Dependent Variable: Positive Number of Breaches	
	(1)	(2)
Post $\times$ Treatment	-0.166*** (0.060)	
SR $\times$ Treatment		-0.100 (0.088)
LR $\times$ Treatment		-0.181*** (0.058)
Observations	1,344	1,344
R^2	0.509	0.509
Period Fixed Effects	Y	Y
Country Fixed Effects	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Large countries are defined as those with a population above the median in 2018.

Table A8: Aggregate Effects: Dropping Brazil and China

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.830*** (0.229)		0.451 (0.461)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.793*** (0.300)		-0.283 (0.577)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.834*** (0.243)		0.546 (0.455)
$\hat{\delta}^{Agg\%}$	-0.564 (0.100)		0.570 (0.723)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.548 (0.136)		-0.246 (0.435)
$\hat{\delta}_{LR}^{Agg\%}$		-0.566 (0.106)		0.726 (0.785)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,660	2,660	2,660	2,660
Pseudo R^2	0.811	0.811	0.885	0.885

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Brazil and China have been removed from the panel. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A9: Aggregate Effects: Excluding COVID Years

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.649*** (0.182)		-0.218 (0.572)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.794*** (0.299)		-0.222 (0.592)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.541*** (0.152)		-0.212 (0.576)
$\hat{\delta}^{Agg\%}$	-0.478 (0.095)		-0.196 (0.460)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.548 (0.135)		-0.199 (0.474)
$\hat{\delta}_{LR}^{Agg\%}$		-0.418 (0.089)		-0.191 (0.466)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	852	852	852	852
Pseudo R^2	0.849	0.849	0.885	0.885

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. All periods after the first quarter of 2020 are excluded from the panel. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A10: Aggregate Effects: Excluding Multinational Organizations

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.915*** (0.280)		0.652 (0.434)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.818*** (0.288)		-0.722 (0.704)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.923*** (0.300)		0.803* (0.443)
$\hat{\delta}^{Agg\%}$	-0.599 (0.112)		0.919 (0.833)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.559 (0.127)		-0.514 (0.342)
$\hat{\delta}_{LR}^{Agg\%}$		-0.603 (0.119)		1.232 (0.989)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,632	2,632	2,632	2,632
Pseudo R^2	0.784	0.784	0.824	0.825

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Data packages originating from multinational organizations are excluded from the panel construction. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\delta^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A11: Alternative Models: Number of Data Breaches

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-0.935*** (0.266)		-0.392*** (0.081)		-0.488*** (0.098)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.797*** (0.297)		-0.350*** (0.106)		-0.439*** (0.129)		-0.123*** (0.032)
LR $\times$ Treatment		-0.947*** (0.284)		-0.401*** (0.077)		-0.500*** (0.093)		-0.105** (0.044)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.692	0.692	0.682	0.682	0.102	0.102
Pseudo R^2	0.793	0.793						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of data breaches per million. Unlike the main specification, the Poisson model does not include a population offset.

Table A12: Alternative Models: Number of Records

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.338 (0.429)		-2.225*** (0.391)		-2.361*** (0.414)		0.949 (20.561)	
SR $\times$ Treatment		-0.223 (0.590)		-1.673*** (0.556)		-1.780*** (0.587)		-0.718 (17.996)
LR $\times$ Treatment		0.402 (0.429)		-2.352*** (0.393)		-2.495*** (0.417)		1.333 (22.551)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.545	0.545	0.542	0.543	0.182	0.182
Pseudo R^2	0.847	0.847						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of records per thousand. Unlike the main specification, the Poisson model does not include a population offset.

Table A13: Alternative Models with Covariates: Number of Data Breaches

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-1.090*** (0.308)		-0.390*** (0.083)		-0.486*** (0.099)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.847*** (0.322)		-0.353*** (0.106)		-0.443*** (0.129)		-0.123*** (0.032)
LR $\times$ Treatment		-1.117*** (0.324)		-0.400*** (0.078)		-0.497*** (0.094)		-0.105** (0.044)
GDP Per Capita	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		
Population	-0.004 (0.006)	-0.005 (0.006)	0.004 (0.003)	0.004 (0.003)	0.006 (0.004)	0.005 (0.004)		
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648	2,648	2,648	2,716	2,716
R^2			0.696	0.696	0.686	0.687	0.102	0.102
Pseudo R^2	0.797	0.797						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of data breaches per million. Annual population data is provided by the World Bank. Unlike the main specification, the Poisson model does not include a population offset.

Table A14: Alternative Models with Covariates: Number of Records

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.478 (0.509)		-2.212*** (0.386)		-2.350*** (0.409)		0.949 (20.561)	
SR $\times$ Treatment		-0.151 (0.618)		-1.721*** (0.550)		-1.831*** (0.581)		-0.718 (17.996)
LR $\times$ Treatment		0.564 (0.520)		-2.343*** (0.389)		-2.489*** (0.413)		1.333 (22.551)
GDP Per Capita	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)		
Population	0.015 (0.015)	0.016 (0.015)	0.020 (0.022)	0.017 (0.022)	0.021 (0.023)	0.017 (0.023)		
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648	2,648	2,648	2,716	2,716
R^2			0.551	0.551	0.549	0.549	0.182	0.182
Pseudo R^2	0.847	0.848						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. In the levels regression, the dependent variable is the number of records per thousand. Annual population data is provided by the World Bank. Unlike the main specification, the Poisson model does not include a population offset.

Table A15: Alternative Models: Number of Data Breaches Scaled by Population

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	-0.927*** (0.266)		-0.090*** (0.020)		-0.110*** (0.026)		-0.109*** (0.041)	
SR $\times$ Treatment		-0.794*** (0.297)		-0.093*** (0.020)		-0.112*** (0.025)		-0.123*** (0.032)
LR $\times$ Treatment		-0.939*** (0.284)		-0.090*** (0.020)		-0.109*** (0.026)		-0.105** (0.044)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.251	0.251	0.230	0.230	0.102	0.102
Pseudo R^2	0.792	0.793						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. The dependent variable is the number of breaches per million, except in the Poisson model, where a log(population) offset is used instead. Annual population data is provided by the World Bank.

Table A16: Alternative Models: Number of Records Scaled by Population

	Poisson		Log(Y + 1)		IHS		Levels	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post $\times$ Treatment	0.342 (0.429)		-0.493*** (0.120)		-0.599*** (0.139)		0.949 (20.561)	
SR $\times$ Treatment		-0.221 (0.590)		-0.412*** (0.152)		-0.502*** (0.178)		-0.718 (17.996)
LR $\times$ Treatment		0.407 (0.429)		-0.511*** (0.124)		-0.621*** (0.143)		1.333 (22.551)
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716	2,716	2,716	2,716	2,716
R^2			0.447	0.447	0.450	0.450	0.182	0.182
Pseudo R^2	0.847	0.847						

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. IHS: inverse hyperbolic sine transformation of the dependent variable. The dependent variable is the number of records per thousand, except in the Poisson model, where a log(population) offset is used instead. Annual population data is provided by the World Bank.

Table A17: Aggregate Effects: No Offset

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.935*** (0.266)		0.338 (0.429)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.797*** (0.297)		-0.223 (0.590)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.947*** (0.284)		0.402 (0.429)
$\hat{\delta}^{Agg\%}$	-0.607 (0.104)		0.402 (0.602)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.549 (0.134)		-0.200 (0.472)
$\hat{\delta}_{LR}^{Agg\%}$		-0.612 (0.110)		0.495 (0.642)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	0.793	0.793	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. No population offset is used. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A18: Aggregate Effects: Weighted Estimation

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.113*** (0.384)		-0.007 (0.473)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.933*** (0.212)		-0.476 (0.558)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.128*** (0.411)		0.040 (0.497)
$\hat{\delta}^{Agg\%}$	-0.671 (0.126)		-0.007 (0.469)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.607 (0.083)		-0.379 (0.347)
$\hat{\delta}_{LR}^{Agg\%}$		-0.676 (0.133)		0.041 (0.517)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	1.426	1.426	1.031	1.031

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. The population offset is removed and observations are instead weighted by population. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A19: Aggregate Effects: Per Capita Outcomes

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.810* (0.447)		0.506 (0.392)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.285 (0.449)		-0.294 (0.709)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.840* (0.467)		0.578 (0.400)
$\hat{\delta}^{Agg\%}$	-0.555 (0.199)		0.658 (0.650)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.248 (0.338)		-0.255 (0.528)
$\hat{\delta}_{LR}^{Agg\%}$		-0.568 (0.201)		0.783 (0.713)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,716	2,716	2,716	2,716
Pseudo R^2	0.071	0.071	0.473	0.474

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A20: Aggregate Effects: Small Countries

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.400*** (0.427)		0.120 (0.436)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.343 (0.674)		-2.253*** (0.626)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.454*** (0.426)		0.262 (0.430)
$\hat{\delta}^{Agg\%}$	-0.753 (0.105)		0.127 (0.491)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.291 (0.478)		-0.895 (0.066)
$\hat{\delta}_{LR}^{Agg\%}$		-0.766 (0.099)		0.300 (0.559)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	1,372	1,372	1,372	1,372
Pseudo R^2	0.473	0.474	0.746	0.751

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are limited to countries with populations below the median population level in 2018. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A21: Aggregate Effects: Large Countries

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-0.863*** (0.283)		0.290 (0.498)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.742** (0.328)		-0.089 (0.617)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-0.874*** (0.300)		0.338 (0.504)
$\hat{\delta}^{Agg\%}$	-0.578 (0.120)		0.336 (0.665)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.524 (0.156)		-0.085 (0.565)
$\hat{\delta}_{LR}^{Agg\%}$		-0.583 (0.125)		0.402 (0.707)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	1,344	1,344	1,344	1,344
Pseudo R^2	0.803	0.803	0.827	0.827

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are limited to countries with populations larger than the median population in 2018. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A22: Aggregate Effects: Size Indicators

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Above Median Pop. x Post	-0.337 (0.360)		-0.528 (0.411)	
Above Median Pop. x SR		0.813* (0.450)		-0.606 (0.584)
Above Median Pop. x LR		-0.334 (0.391)		-0.461 (0.360)
Post x Treatment	-1.399*** (0.423)		0.111 (0.435)	
SR x Treatment		-0.323 (0.674)		-2.313*** (0.603)
LR x Treatment		-1.337** (0.540)		-0.031 (0.458)
Above Median Pop. x Post x Treatment	0.536 (0.507)		0.178 (0.657)	
Above Median Pop. x SR x Treatment		-0.477 (0.745)		2.265*** (0.855)
Above Median Pop. x LR x Treatment		0.273 (0.558)		0.503 (0.839)
Observations	2,716	2,648	2,716	2,648
Pseudo R^2	0.793	0.797	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are unweighted.

Table A23: Aggregate Effects: With Covariates

	Number of Breaches		Number of Records	
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{Agg}$)	-1.051*** (0.285)		0.392 (0.471)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{Agg}$)		-0.834*** (0.320)		-0.190 (0.596)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{Agg}$)		-1.072*** (0.298)		0.465 (0.482)
GDP Per Capita	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
$\hat{\delta}^{Agg\%}$	-0.650 (0.100)		0.480 (0.697)	
$\hat{\delta}_{SR}^{Agg\%}$		-0.566 (0.139)		-0.173 (0.493)
$\hat{\delta}_{LR}^{Agg\%}$		-0.658 (0.102)		0.591 (0.768)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	2,648	2,648	2,648	2,648
Pseudo R^2	0.796	0.797	0.847	0.847

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Observations are unweighted. $\hat{\delta}^{Agg\%}$ terms are calculated by transforming the $\hat{\delta}^{Agg}$ terms. $\hat{\delta}^{Agg\%} = \exp(\hat{\delta}) - 1$. The standard errors for these terms are calculated using the Delta Method: $SE(\hat{\delta}^{Agg\%}) = \exp(\hat{\delta}^{Agg})SE(\hat{\delta}^{Agg})$.

Table A24: Data Package Effects: PII Fraction - Excluding Emails and Passwords

	Dependent Variable: PII Fraction			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.004 (0.020)		-0.013 (0.014)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.014 (0.042)		-0.004 (0.042)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.003 (0.022)		-0.014 (0.016)
Multinational			0.051*** (0.019)	0.051*** (0.019)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.422	0.422	0.423	0.423

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. PII definition excludes emails and passwords.

Table A25: Data Package Effects: Number of PII Records - Excluding Emails and Passwords

	Dependent Variable: Log(Number of PII Records)			
	(1)	(2)	(3)	(4)
Post x Treatment	0.948 (0.638)		0.364 (0.390)	
SR x Treatment		1.846** (0.824)		1.383** (0.680)
LR x Treatment		0.897 (0.591)		0.369 (0.402)
Multinational			1.809*** (0.346)	1.741*** (0.337)
$\hat{\delta}$	1.579 (1.647)		0.439 (0.562)	
$\hat{\delta}^{SR}$		5.337 (5.224)		2.987 (2.710)
$\hat{\delta}^{LR}$		1.451 (1.448)		0.447 (0.582)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	4,394	4,394	4,394	4,394
R^2	0.382	0.383	0.386	0.386

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. PII definition excludes emails and passwords.

Table A26: Data Package Effects: Number of Records—Full Sample

	Dependent Variable: Log(Number of Records)			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.956** (0.400)		0.563** (0.254)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.657 (0.462)		0.221 (0.297)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.989** (0.405)		0.599** (0.266)
Multinational			1.424*** (0.308)	1.426*** (0.308)
$\hat{\delta}^{DP\%}$	1.602 (1.041)		0.756 (0.445)	
$\hat{\delta}_{SR}^{DP\%}$		0.929 (0.891)		0.248 (0.371)
$\hat{\delta}_{LR}^{DP\%}$		1.687 (1.089)		0.821 (0.484)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.280	0.280	0.288	0.289

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table A27: Data Package Effects: PII Fraction—Full Sample

	Dependent Variable: PII Fraction			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	-0.003 (0.007)		-0.008 (0.009)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		-0.019 (0.030)		-0.025 (0.025)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		-0.001 (0.007)		-0.007 (0.011)
Multinational			0.020 (0.015)	0.020 (0.015)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.402	0.402	0.403	0.403

** $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$*

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table A28: Data Package Effects: Number of Data Types—Full Sample

	Dependent Variable: Number of Unique Data Types			
	(1)	(2)	(3)	(4)
Post $\times$ Treatment ($\hat{\delta}^{DP}$)	0.524** (0.249)		0.527** (0.243)	
SR $\times$ Treatment ($\hat{\delta}_{SR}^{DP}$)		0.263 (0.549)		0.266 (0.576)
LR $\times$ Treatment ($\hat{\delta}_{LR}^{DP}$)		0.552* (0.291)		0.554* (0.280)
Multinational			-0.011 (0.207)	-0.010 (0.207)
Period Fixed Effects	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y
Observations	5,669	5,669	5,669	5,669
R^2	0.277	0.277	0.277	0.277

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. Estimates use the full sample and do not drop early period data packages.

Table A29: Data Types Extensive Margin Effects

	Account		Email		Financial		Passwords		PII	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Post $\times$ Treatment	0.085** (0.037)		0.033 (0.034)		-0.011 (0.012)		0.033 (0.061)		0.014 (0.027)	
SR $\times$ Treatment		0.087 (0.072)		0.061*** (0.016)		0.011 (0.027)		0.135*** (0.030)		-0.028 (0.065)
LR $\times$ Treatment		0.085* (0.044)		0.030 (0.040)		-0.014 (0.013)		0.023 (0.069)		0.019 (0.031)
Observations	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394	4,394
R^2	0.275	0.275	0.378	0.378	0.067	0.067	0.358	0.359	0.477	0.477
Period Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Country Fixed Effects	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Note: Standard errors are clustered at the country level. The extensive margin is estimated using a linear probability model. The dependent variable is an indicator for whether the data package contains data of each type. PII in columns 9 and 10 does not include emails or passwords.